SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks50** **Duration :60 Minutes**

Annexure A

Analytical Problem Solving

Code of Conduct:

I Bharath sankar(192425046) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: bharath s

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	

1. Grey Level Transformation

(a) Explain the purpose of grey level transformation in image enhancement.
Grey level transformation is used to **improve the appearance of an image** by changing the brightness and contrast of pixel values. It helps highlight important details and suppress unwanted information.

(b) Apply a negative transformation s=255-r to the pixel values:
r=[50,100,150,200]
Compute the transformed values.
Formula:

$$s = 255 - r$$

Original Pixel (r)	Transformed Pixel (s)
50	205
100	155
150	105
200	55

Answer:

$$\boxed{[205, \ ,155, \ 105 \ 55]}$$

2. Log Transformation

(a) Explain how log transformation enhances low-intensity pixels.
Log transformation **expands low-intensity (dark) pixel values** and compresses high-intensity values, making dark regions more visible.

(b) Given s=clog(1+r), where c=10, compute s for:
r=[0,10,50,100]
Formula:

$$s = c\log(1 + r)$$

Given:

$$c = 10$$

Using natural logarithm (ln):

r	$10\ln(1+r)$	**s (Approx.)**
0	$10\ln 1$	0.00
10	$10\ln 11$	23.98
50	$10\ln 51$	39.32
100	$10\ln 101$	46.15

Answer:

$$\boxed{[0, \ ,23.98, \ 39.32 \ 46.15]}$$

3. Histogram Processing
(a) Explain the role of histogram equalization in contrast enhancement.
Histogram equalization **improves image contrast** by redistributing pixel intensities over the available grey levels.

(b) Given a 3-bit image with histogram:

Intensity	**Frequency**
0	2
1	2
2	4
3	8

Compute the cumulative distribution and equalized intensity values.
Given Histogram:

Intensity	**Frequency**
0	2
1	2
2	4
3	8

Total pixels:

$$N = 2 + 2 + 4 + 8 = 16$$

Step 1: Cumulative Distribution

Intensity	Frequency	Cumulative Frequency
0	2	2
1	2	4
2	4	8
3	8	16

Step 2: Equalized Intensity
For a 3-bit image,

$$L = 8$$

Formula:

$$s = (L - 1) \times \frac{\text{CDF}}{N}$$
$$L - 1 = 7$$

Intensity	**CDF**	**Equalized Value**
0	2	$7 \times \frac{2}{16} = 0.875 \approx 1$
1	4	$7 \times \frac{4}{16} = 1.75 \approx 2$
2	8	$7 \times \frac{8}{16} = 3.5 \approx 4$

Intensity	CDF	Equalized Value
3	16	$7 \times \frac{16}{16} = 7$

Answer:

Original	Equalized
0	1
1	2
2	4
3	7

4. Spatial Filtering (Smoothing)

(a) Explain the purpose of spatial filtering for smoothing.

Smoothing removes **noise** and reduces small intensity variations by averaging neighbouring pixels.

(b) Apply a 3×3 averaging filter to the following region:

10 20 30 20 30 40 30 40 50

Compute the new center pixel value.

Matrix:

$$\begin{matrix} 10 & 20 & 30 \\ 20 & 30 & 40 \\ 30 & 40 & 50 \end{matrix}$$

Sum:

$$10 + 20 + 30 + 20 + 30 + 40 + 30 + 40 + 50 = 270$$

Average:

$$\frac{270}{9} = 30$$

New centre pixel

$$\boxed{30}$$

5. Spatial Filtering (Sharpening)

(a) Explain how sharpening enhances image details.

Sharpening enhances **edges and fine details** by emphasizing rapid intensity changes.

(b) Apply the Laplacian mask:

[0 -1 0}, {-1 4 -1], { 0 -1 0]

to the matrix:

10 10 10, 10 20 10, 10 10 10

Compute the output at the center.

Mask:

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

Image:

$$\begin{matrix} 10 & 10 & 10 \\ 10 & 20 & 10 \\ 10 & 10 & 10 \end{matrix}$$

Calculation:

$$(4 \times 20) - (10 + 10 + 10 + 10)$$
$$80 - 40 = 40$$

Output at the centre

$$\boxed{40}$$